9. Travelling Salesman Problem (TSP)

Aim:

To find the minimum-cost tour for visiting all cities and returning to the starting city.

Algorithm:

1. Define the cost matrix for all cities.

2. Fix the first city as the starting point.

3. Generate all possible orders of the remaining cities.

4. Calculate the total travel cost of each tour.

5. Select and display the tour with the minimum cost.

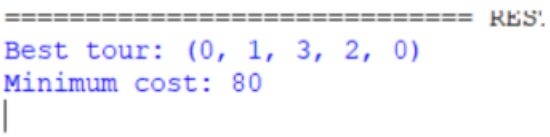
Program:

```
from itertools import permutations
cost = [
[0, 10, 15, 20],
[10, 0, 35, 25],
[15, 35, 0, 30],
[20, 25, 30, 0]
]
n = len(cost)
best_cost = float('inf')
best_path = []
for p in permutations (range (1, n)):
path = (0,) + p + (0,)
total = 0
for i in range (len (path) - 1):
total += cost[path[i]][path[i + 1]]
if total < best_cost:
best_cost = total


best_path = path
```

```
print ("Best tour:", best_path)
print ("Minimum cost:", best_cost)
```

Output:



Result:

Thus, the Travelling Salesman Problem was solved successfully using Python.